Aggregate output, Y, and the total number of hours worked, N, are given by

$$Y = (1-u)\int_{\mu} f[x, n(x), k]dH(x),$$

and $N = (1-u)\int_{\mu} n(x)dH(x)$, with $\mu \equiv \max(R, \phi)$. Using (1) and (8) gives

$$Y(K_e, \mu) = [1 - H(\mu)]K_e E_H(x \mid x \geq \mu) \tag{18}$$

$$N = [1 - H(\mu)]K_e, \tag{19}$$

where $E_H(x \mid x \geq \mu) = [1 - H(\mu)]^{-1}\int_{\mu} x dH(x)$. Intuitively, since every firm–worker pair is setting hours either to 0 or to full capacity k, the aggregate number of hours worked is just equal to the fraction of firm–worker pairs who engage in production times the total capital stock in filled jobs. Similarly, aggregate output equals the number of *active* units of capital, $[1 - H(\mu)]K_e$, weighted by their average productivity.[18] Following Houthakker (1955–1956), one could imagine solving (19) for the "aggregate labour demand" by active firms, $\mu(K_e, N)$, and then substituting it in (18) to obtain $Y[K_e, \mu(K_e, N)]$. Hereafter, I use $F(K_e, N)$ to denote $Y[K_e, \mu(K_e, N)]$ to simplify notation and to stress the fact that this is the economy's *aggregate production function.* This aggregate production function exhibits constant returns to scale even for an arbitrary H. To see this, notice that $\mu(K_e, N)$ is homogeneous of degree 0, and hence (18) indicates that for any $\zeta > 0$, $F(\zeta K_e, \zeta N) = \zeta F(K_e, N)$.[19]

Suppose idiosyncratic shocks are draws from a Pareto distribution

$$G(x) = \begin{cases} 0 & \text{if } x < \varepsilon \\ 1 - \left(\frac{\varepsilon}{x}\right)^{\alpha} & \text{if } \varepsilon \leq x, \end{cases} \tag{20}$$

where $\varepsilon > 0$ and $\alpha > 1$. Then, provided $R \geq \varepsilon$, $1 - G(R) = \left(\frac{\varepsilon}{R}\right)^{\alpha}$, and hence $G(x) - G(R) = (\varepsilon/R)^{\alpha}[1 - (R/x)^{\alpha}]$ for any $x \geq R$. Substituting these expressions in (3) yields

$$H(x) = \begin{cases} 0 & \text{if } x < R \\ 1 - \left(\frac{R}{x}\right)^{\alpha} & \text{if } R \leq x. \end{cases} \tag{21}$$

So the steady-state productivity distribution of active matches follows a Pareto distribution with parameters R and α. Using (21), $1 - H(\mu) = \left(\frac{R}{\mu}\right)^{\alpha}$, and $E_H(x \mid x \geq \mu) = \frac{\alpha}{\alpha-1}\mu$, so the aggregates (18) and (19) specialize to $Y(K_e, \mu) = \frac{\alpha}{\alpha-1}R^{\alpha}\mu^{1-\alpha}K_e$, and $N = (R/\mu)^{\alpha}K_e$. Inverting the latter yields $\mu = (K_e/N)^{1/\alpha}R$, and substituting this in the former implies

$$F(K_e, N) = AK_e^{\gamma}N^{1-\gamma}, \tag{22}$$

18. As mentioned in footnote 8, Mortensen and Pissarides (1994) assume that G has support [0, 1] and that all new matches start off with productivity 1. So, with $\delta = 0$, aggregate output in their model evolves according to

$$\dot{Y} = \theta q(\theta)uk - \lambda Y + \lambda(1-u)k\int_{\mu}^{1} x dG(x).$$

Replacing $(1-u)k$ with K_e, steady-state output is $Y = \frac{\theta q(\theta)uk}{\lambda} + [1 - H(\mu)]K_e E_H(x \mid x \geq \mu)$, which looks like (18) except for the first term. Assuming that the initial productivity of a new match is a random draw from G—just as the innovations to the productivity of ongoing matches—allows for a density G with unbounded support. In addition, this alternative assumption smooths aggregate output by getting rid of the "spike" $\theta q(\theta)uk\lambda^{-1}$.

19. Also, from (18), $F_2(K_e, N) = -\mu_2(K_e, N)K_e\mu dH(\mu)$, and from (19), $-\mu_2(K_e, N)K_e dH(\mu) = 1$. Thus, $F_2(K_e, N) = \mu$. So the marginal product of labour in the aggregate production function is equal to the marginal product of the least efficient unit of labour employed in production. I owe this argument to Erzo G. J. Luttmer.

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where

$$A = \frac{R}{1-\gamma} \tag{23}$$

and $\gamma \equiv 1/\alpha$. This aggregation result is reminiscent of the classic aggregation result of Houthakker (1955–1956).[20] The factor A is what macroeconomists normally refer to as TFP. Its level depends on α, a parameter of the primitive distribution of productivity shocks, as well as on all the characteristics of the labour market as summarized by the destruction decision, R.

5. LABOUR-MARKET POLICIES AND THE LEVEL OF TFP

This section considers the effects of four policies: employment subsidies, hiring subsidies, firing taxes, and unemployment benefits. I follow Pissarides (2000) and model the subsidies as transfers from the government to the firm and the firing taxes as payments from the firm to the government.[21] The value function $W(x)$ is still given by (5), while (4), (6), and (7) generalize to

$$rU = \tau_b + \theta q(\theta) \int \max[W_o(z) - U, 0] dG(z)$$

$$rV = -ck + q(\theta) \int \max[J_o(z) + \tau_h k - V, 0] dG(z)$$

$$rJ(x) = \pi(x) + \tau_e k + \lambda \int \max[J(z) - V + \tau_f k, 0] dG(z) - (\delta + \lambda)[J(x) - V + \tau_f k].$$

The policy variables are a hiring subsidy, τ_h, an employment subsidy, τ_e, a firing tax, τ_f, and unemployment benefits, τ_b.[22] In Appendix B I show that if G is as in (20), the job-destruction and job-creation conditions that determine the equilibrium pair (θ, R) are

$$R - \phi - c + \tau_e + r\tau_f - \left(\tau_b + \frac{\beta}{1-\beta} c\theta\right) + \frac{\lambda \varepsilon^\alpha R^{1-\alpha}}{(\alpha - 1)(r + \delta + \lambda)} = 0 \tag{24}$$

$$\frac{\varepsilon^\alpha R^{1-\alpha}}{(\alpha - 1)(r + \delta + \lambda)} + \left(\frac{\varepsilon}{R}\right)^\alpha (\tau_h - \tau_f) - \frac{c}{(1-\beta) q(\theta)} = 0. \tag{25}$$

The main properties of the equilibrium are summarized as follows.

Proposition 1. *Suppose* $\varepsilon + (\alpha - 1)(r + \delta + \lambda)(\tau_h - \tau_f) > 0$. *Let* θ_ϵ^* *be defined by* $q(\theta_\epsilon^*) = \frac{(\alpha-1)(r+\delta+\lambda)c}{(1-\beta)[\varepsilon+(\alpha-1)(r+\delta+\lambda)(\tau_h-\tau_f)]}$, *and* $\phi_\epsilon = \left[1 + \frac{\lambda}{(\alpha-1)(r+\delta+\lambda)}\right]\varepsilon - \left(1 + \frac{\beta}{1-\beta}\theta_\epsilon^*\right)c + \tau_h + r\tau_f - \tau_b$. *If* $\varepsilon + \alpha(r + \delta + \lambda)(\tau_h - \tau_f) > 0$, *then for any* $\phi > \phi_\varepsilon$ (a) *there exists a unique equilibrium,* (b) $R > \varepsilon$, (c) $\partial R/\partial\phi > 0$, *and* (d) $\partial\theta/\partial\phi < 0$. *If in addition* $\phi_\epsilon - \varepsilon > 0$, *then* (e) *there is a nondegenerate interval* $(\phi_\epsilon, \widetilde{\phi})$ *such that* $R(\phi) < \phi$ *for all* $\phi \in (\phi_\epsilon, \widetilde{\phi})$.

Aggregate output is still given by (22); the aggregate stock of capital in filled jobs, K_e, is still given by (17); and the aggregate number of hours worked, N, is still given by $N = (R/\mu)^\alpha K_e$. In addition, if the measure of capital used to construct aggregate output is K_e, then the level of TFP

20. I discuss this relationship at some length in Section 6.

21. I assume that upon separation, the firm must pay the firing tax to the government because in the present set-up, firing taxes would be completely neutral under the alternative scheme in which the firm compensates the fired worker directly.

22. All payments are assumed to be proportional to the firm's size, as measured by k. For example, upon separation, the firm is taxed $\tau_f k$, and while unemployed, the worker receives $\tau_b k$ from the government. This assumption, borrowed from Pissarides (2000), is convenient because it ensures that policies introduce no scale effects into the job-creation and job-destruction decisions. Also, to keep the analysis simple, I will ignore the government's financing constraints. A natural extension would be requiring the government to run a balanced budget. A simple example of a scheme that is self-financing in the steady state is $\tau_f = \tau_h$, and $\tau_b = \tau_e = 0$.

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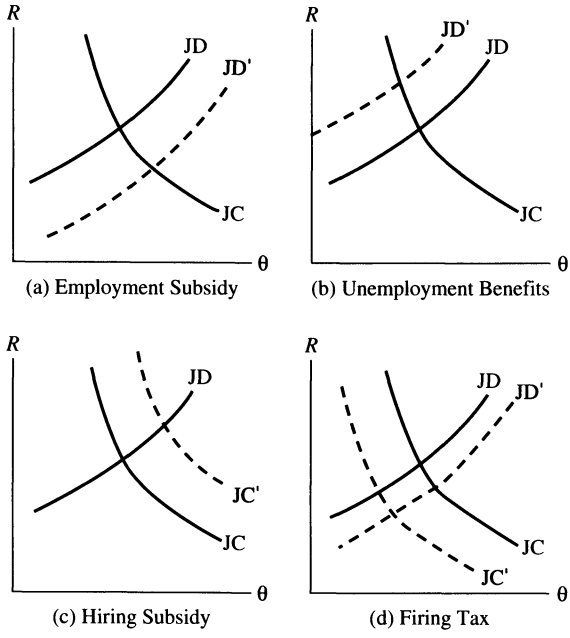

FIGURE 1

Equilibrium effects of policies on job-creation (JC) and job-destruction (JD) rates

is still as in (23). The following result, which holds under the assumptions stated in Proposition 1, summarizes the effects that labour-market policies have on the level of TFP.

Proposition 2. *Employment subsidies and firing taxes reduce A. Hiring subsidies and unemployment benefits increase A.*

Since A is proportional to R, policies have the same qualitative effect on TFP as they have on the destruction rate. Proposition 2 is illustrated in Figure 1. In each panel, the dashed lines labelled JC′ and JD′ indicate how the curve shifts in response to the introduction of the policy.

Employment subsidies make firms more tolerant of low productivity realizations and hence lower the average productivity of active firms. All else equal, an economy with relatively high subsidies to continued employment will exhibit a low job-destruction rate, a high job-creation rate, and hence low levels of unemployment and measured TFP. Firing taxes have a similar qualitative effect on job destruction, but that mechanism is reinforced by a relatively low rate of job creation (which reduces the reservation wage and hence makes firms even more tolerant of low productivity realizations). So firing restrictions will reduce measured TFP, as well as the job-creation and job-destruction rates. Hiring subsidies have no direct effect on the destruction decision, but they do stimulate job creation. This increases market tightness, which in turn increases the worker's outside option and raises measured TFP, job creation, and job destruction. Unemployment benefits also cause R to rise through an increase in the worker's reservation wage. Consequently, economies with relatively high unemployment benefits will tend to exhibit relatively high levels of TFP and unemployment.[23]

23. Notice that these results obtain if factor inputs are measured rather accurately, so that the TFP residual corresponds to (23). But naturally, what is measured as the TFP residual, as well as how it responds to policy, *will* depend on the accuracy with which factor inputs are measured. See Sections 6.2 and 7 below.

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6. DISCUSSION

In this section I elaborate on several issues that were raised in the previous sections. First, I explain the similarities and differences between the well-known aggregation result of Houthakker (1955–1956) and the one I obtained in Section 4. I then use the aggregation result to show how the observed level of TFP depends on the measurement of factor inputs. Finally, I comment on the implications of the theory for the standard growth accounting exercises that presume frictionless markets and competitive pricing.

6.1. *Houthakker*

Houthakker (1955–1956) studies the problem of aggregating the technological possibilities of individual production units (*e.g.* firms) into a production function for a group of units (*e.g.* an industry). He considers a set of heterogeneous production units that can employ two variable factors, x_1 and x_2, to produce output, x_0, and face capacity constraints due to a fixed, unmodelled factor. Each unit j has access to a fixed-proportions technology $x_{0j} = \min\left(\frac{x_1}{a_{1j}}, \frac{x_2}{a_{2j}}, \varphi_j\right)$. That is, a_{1j} units of input 1 and a_{2j} units of input 2 are needed to produce one unit of x_0, and the unmodelled fixed inputs limit the production unit's possible output to φ_j units of x_0. The set-up is static and the notion of equilibrium partial: each production unit j seeks to maximize profit, $1 - x_{1j}p_1 - x_{2j}p_2$, where p_i is the price of input $i = 1, 2$ (the price of output has been normalized to 1). The optimal choice of inputs for production unit j is

$$x_{ij} = \begin{cases} a_{ij}\varphi_j & \text{if } 1 - a_{1j}p_1 - a_{2j}p_2 \geq 0 \\ 0 & \text{if } 1 - a_{1j}p_1 - a_{2j}p_2 < 0, \end{cases} \tag{26}$$

for $i = 1, 2$. Units that can attain positive profit produce at full capacity, that is, they set $x_{0j} = \varphi_j$, whereas those that cannot, remain inactive, that is, they set $x_{ij} = 0$ for $i = 0, 1, 2$. Let $a = (a_1, a_2)$ and assume a density, $\varphi(a)$, of (the various levels of) capacity constraints across production cells. Observe that total output is

$$X_0 = f_0(p_1, p_2), \tag{27}$$

where $f_0(p_1, p_2) \equiv \int_0^{p_1^{-1}} \int_0^{\frac{1+a_1p_1}{p_2}} \varphi(a)da$. The total amount used of variable input i is

$$X_i = f_i(p_1, p_2) \quad \text{for } i = 1, 2, \tag{28}$$

where $f_i(p_1, p_2) \equiv \int_0^{p_1^{-1}} \int_0^{\frac{1+a_1p_1}{p_2}} a_i\varphi(a)da$. Then, similar to what was done in Section 4, one may be able to solve the two expressions in (28) for prices, that is, $p_i = g_i(X_1, X_2)$ and then substitute these expressions back into (27) to get a functional relationship between aggregate output and the aggregate quantities of the variable inputs: the aggregate production function. Specifically, $X_0 = F(X_1, X_2)$, where $F(X_1, X_2) \equiv f_0[g_1(X_1, X_2), g_2(X_1, X_2)]$. Clearly, the shape of the aggregate F will depend on the particular shape of the distribution of capacities, φ. Houthakker's key insight was that if φ is of the "generalized Pareto type", that is, $\varphi(a) = \kappa a_1^{\alpha_1 - 1} a_2^{\alpha_2 - 1}$, with $\alpha_1 \geq 1$, $\alpha_2 \geq 1$, and $\kappa \geq 0$, then $F(X_1, X_2) = C X_1^{\gamma_1} X_2^{\gamma_2}$, where C, γ_1, and γ_2 are expressions involving only the parameters of the primitive distribution (α_1, α_2, and κ), and $\gamma_1 + \gamma_2 < 1$.

In my analysis, I have assumed each production unit employs a single variable factor (labour) as well as capital. Capital is chosen before engaging in search and then remains fixed, hence playing the role of the fixed factor constraining output at the time that employment and production decisions are made. In Houthakker's language, I am working with a single variable production input and explicitly modelling the fixed factor that constrains capacity. This formulation delivers an aggregate production function with constant returns to scale, whereas the set-up used by

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